E. Bear and Bad Powers of 42

time limit per test: 5 seconds
memory limit per test: 256 megabytes

Limak, a bear, isn't good at handling queries. So, he asks you to do it.

We say that powers of 42 (numbers $1, 42, 1764, ...$) are *bad*. Other numbers are *good*.

You are given a sequence of n good integers $t_1, t_2, ..., t_n$. Your task is to handle q queries of three types:

1. `1 i` — print t_i in a separate line.
2. `2 a b x` — for $i \in [a, b]$ set t_i to x. It's guaranteed that x is a good number.
3. `3 a b x` — for $i \in [a, b]$ increase t_i by x. After this repeat the process while at least one t_i is bad.

You can note that after each query all t_i are good.

Input

The first line of the input contains two integers n and q ($1 \le n, q \le 100\,000$) — the size of Limak's sequence and the number of queries, respectively.

The second line of the input contains n integers $t_1, t_2, ..., t_n$ ($2 \le t_i \le 10^9$) — initial elements of Limak's sequence. All t_i are good.

Then, q lines follow. The i-th of them describes the i-th query. The first number in the line is an integer $type_i$ ($1 \le type_i \le 3$) — the type of the query. There is at least one query of the first type, so the output won't be empty.

In queries of the second and the third type there is $1 \le a \le b \le n$.

In queries of the second type an integer x ($2 \le x \le 10^9$) is guaranteed to be good.

In queries of the third type an integer x ($1 \le x \le 10^9$) may be bad.

Output

For each query of the first type, print the answer in a separate line.

Example

input Copy

```
6 12
40 1700 7 1672 4 1722
3 2 4 42
1 2
1 3
3 2 6 50
1 2
1 4
1 6
2 3 4 41
3 1 5 1
1 1
1 3
1 5
```

output Copy

```
1742
49
1842
1814
1822
43
44
107
```

Note

After a query `3 2 4 42` the sequence is $40, 1742, 49, 1714, 4, 1722$.

After a query `3 2 6 50` the sequence is $40, 1842, 149, 1814, 104, 1822$.

After a query `2 3 4 41` the sequence is $40, 1842, 41, 41, 104, 1822$.

After a query `3 1 5 1` the sequence is $43, 1845, 44, 44, 107, 1822$.

Codeforces Round 356 (Div. 1)

Finished

Practice

→ **Virtual participation**

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Start virtual contest

→ **Clone Contest to Mashup**

You can clone this contest to a mashup.

Clone Contest

→ **Submit?**

Language: GNU G++23 14.2 (64 bit, ms

Choose file: Choose File No file chosen

Submit

→ **Last submissions**

Submission	Time	Verdict
393705547	Oct/08/2026 14:26	Accepted
393705324	Oct/08/2026 14:24	Wrong answer on test 1
393700853	Oct/08/2026 13:44	Wrong answer on test 1

→ **Problem tags**

data structures *3100

No tag edit access

→ **Contest materials**

- Announcement (en)
- Tutorial (en)